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CS 3151 – Software Engineering

Requirements Engineering Report

Phase 1 – Midterm Submission

Tempo – AI Marketing Co-Pilot

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Abstract

Tempo is an AI-powered, bilingual (Arabic/English) marketing co-pilot for solo marketers, SMEs, freelance creators and boutique agencies in Saudi Arabia. It combines Saudi-dialect AI content, scheduling to Snapchat, X and Instagram, unified social and website analytics and automated bilingual reports in one SAR-priced dashboard. This report is the final version of the project requirements (analysis phase). It brings together the four labs — Lab 1 Market Study, Lab 2 Personas & User Stories, Lab 3 Use Cases & UML and Lab 4 SRS — revised so that every requirement is SMART.

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1 Introduction

1.1 Project Overview

Develop Tempo, an AI-powered marketing co-pilot that helps solo marketers, SMEs, freelance content creators and agencies in Saudi Arabia manage social media and website marketing from one bilingual (Arabic/English) dashboard. Tempo replaces a fragmented daily routine — logging into Snapchat, X, Instagram and Google Search Console separately, rewriting English-first AI output into Saudi Arabic, and building reports by hand — with a single workflow: plan and draft, schedule, measure, report.

Features include Saudi-dialect AI content generation (Najdi and Hijazi as well as Modern Standard Arabic and English), a unified content calendar with scheduled publishing, a unified social and website analytics dashboard, automated weekly bilingual reports with branded PDF export, Saudi-native timing and trend insights, client workspaces and role-based team access for agencies, and SAR-denominated pricing with a freemium entry point.

1.2 Problem to Be Solved

Saudi Arabia has more than 25 million active social media users and a fast-growing e-commerce sector driven by Vision 2030, yet the tools its marketers rely on were built for Western markets. The market study (survey of 47 Saudi marketing professionals in Riyadh and Jeddah, Q3 2026) identified three gaps:

- **Weak Arabic AI:** existing AI writers produce generic or formal Arabic that must be rewritten into Saudi dialect, costing about 4 hours per week.
- **Fragmented platforms:** no tool unifies Snapchat, X, Instagram and Google Search Console in a single Arabic-language dashboard, so data is consolidated manually (6 hours per week for a solo marketer and 12+ hours per week for an agency team).
- **Unaffordable, USD-priced tools:** enterprise suites cost more than SAR 900 per month and foreign-currency pricing creates unpredictable costs for SMEs.

Problem statement. Saudi marketers and small agencies lose between 4 and 12 hours every week rewriting AI output into Saudi dialect, switching between four platforms and building reports by hand, because no affordable, SAR-priced tool combines Saudi-dialect content, scheduling for the Saudi platform mix and unified analytics.

1.3 Objectives

The project objectives are written in SMART form (each has a measurable target and a date):

- Generate three ready-to-edit Saudi-dialect (Najdi, Hijazi), MSA or English drafts from a short brief in under 10 seconds, in the MVP released by February 2027.
- Schedule and publish content to X and Instagram from one unified calendar by the public launch (February 2027), and to Snapchat by April 2027 (subject to platform approval).
- Combine social metrics and Google Search Console website data in one bilingual dashboard by February 2027.
- Reduce weekly reporting time for a solo marketer from about 6 hours to under 1 hour through automated bilingual reports, measured with closed-beta users (January–February 2027).
- Support agencies with client workspaces, role-based access control and branded PDF reports by June 2027.
- Offer SAR pricing (Starter at SAR 189/month, Agency at SAR 499/month) with a free trial from the public launch, and reach 150 paying accounts by the end of 2027.
- Comply with the Saudi Personal Data Protection Law (PDPL) and host all customer data in a Saudi cloud region from day one of the closed beta.

1.4 Scope

In scope (three-year plan): a mobile-responsive web application (progressive web app) for solo marketers, SMEs, freelance creators and boutique agencies in Riyadh, Jeddah and Dammam, followed by a GCC pilot. **Out of scope:** native mobile apps for the MVP, large in-house enterprise marketing departments (identified as a user group but deferred), paid-advertising management, and training a proprietary language model.

2 SDGs and Vision 2030

2.1 Sustainable Development Goals

- **SDG 8 (Decent Work and Economic Growth):** supports economic growth and digital jobs by saving SMEs and freelancers about 4–11 hours of manual work per week.
- **SDG 9 (Industry, Innovation and Infrastructure):** brings local, AI-powered innovation and Arabic-language AI to small Saudi businesses.
- **SDG 10 (Reduced Inequalities):** affordable SAR pricing and a free tier give SMEs and freelancers tools that were only affordable for large companies.
- **SDG 17 (Partnerships for the Goals):** connects Saudi businesses to global platforms (Snapchat, X, Instagram and Google).

2.2 Saudi Vision 2030

- **Digital Economy:** grows the digital economy's contribution to Saudi GDP by moving marketing from spreadsheets to an automated platform.
- **SME Empowerment:** tools built for small businesses, at an SME price.
- **Data Sovereignty:** customer data stored in a Saudi cloud region (AWS KSA), in line with the PDPL.
- **Local Entrepreneurship:** a Saudi-built product that writes in Saudi dialects and follows local occasions (Ramadan, Eid, National Day, Founding Day).

3 Market Study (Lab 1)

3.1 Saudi Digital Marketing Environment

Saudi Arabia is home to more than 25 million active social media users and a rapidly expanding e-commerce sector driven by Vision 2030. Snapchat, X and Instagram dominate how Saudi consumers discover brands, but the available tools were designed elsewhere: they lack Arabic-language AI, do not integrate the Saudi-dominant platforms with website analytics, and are priced in foreign currency.

3.2 Market Size

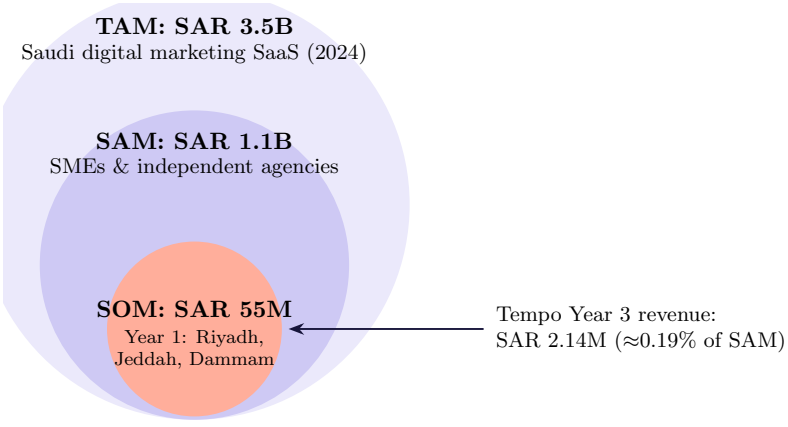


Figure 1: Market Size (TAM / SAM / SOM)

The Saudi digital marketing SaaS market is worth SAR 3.5 billion (TAM, 2024), with a serviceable addressable market of SAR 1.1 billion (SMEs and independent agencies) and a Year 1 obtainable market of SAR 55 million. The SOM is the market study's estimate of what is reachable, not a sales forecast; Tempo's own revenue plan is deliberately much smaller (Year 3 revenue of SAR 2.14M is about 0.19% of the SAM).

3.3 Target Customer Segments

Segment	Profile	Key pain point	Use
Solo marketers & SME owners	Manage own social and website marketing (persona: Nora)	Four separate platforms; English-first AI; USD pricing	Daily
Boutique agencies	Manage multiple client accounts (persona: Khalid)	12+ hours/week compiling data; no affordable multi-client workspaces; formal Arabic instead of client dialect	Daily
Freelance content creators	Produce Arabic content for several brands (persona: Lama)	Slow drafting and rewriting; one calendar per brand	Several times a week

Table 1: Target Segments (initial focus: Riyadh, Jeddah and Dammam, then GCC)

3.4 Competitor Analysis

Four global tools lead the market, but none of them was built for Saudi marketers. Table 2 compares them with Tempo feature by feature and by monthly price.

	Tempo	Hootsuite	Buffer	Sprout Social	Jasper AI
Price per month	**SAR 189**	SAR 370	SAR 56	SAR 934	SAR 184
Saudi dialect AI	✓ Najdi, Hijazi, MSA, English	✗ None	✗ None	✗ None	✗ Formal Arabic only
Arabic content	✓ Arabic + English	✗ No	✗ No	✓ Bilingual posting	✓ Arabic copy
Snapchat	✓ Publish + analytics	✗ No	✗ No	– Not confirmed	✗ No
Scheduling	✓ Snapchat, X, Instagram	✓ Yes	✓ Yes	✓ Yes	✗ No
Website analytics (Search Console)	✓ In the same dashboard	✗ No	✗ No	✗ No	✗ No
Reports	✓ Weekly, Arabic + English	○ Basic	○ Basic	✓ Enterprise	✗ No
Saudi insights	✓ Saudi times, Eid, National Day	✗ No	✗ No	✗ No	✗ No
Currency	✓ SAR (no FX risk)	✗ USD	✗ USD	✗ USD	✗ USD
Main gap	—	No Arabic AI, no Snapchat	Scheduling only	Too expensive for SMEs	Writing only, no publishing

Table 2: Competitor Comparison (✓ = yes, ✗ = no, ○ = basic only)

What this shows:

- **Tempo is the only tool with Saudi dialect AI** (Najdi and Hijazi). The others write in English or formal Arabic, so marketers spend about 4 hours a week rewriting.
- **Tempo is cheaper than the all-in-one tools**: SAR 189 against SAR 370 for Hootsuite (49% cheaper) and SAR 934 for Sprout Social (80% cheaper).
- **Buffer and Jasper are cheaper, but each does only one job**: Buffer only schedules and Jasper only writes. Using both costs SAR 240 a month — more than Tempo — and still gives no Saudi dialect, no Snapchat and no website analytics.
- **Tempo is the only one priced in SAR**; the others charge in USD, so the cost changes with the exchange rate.

3.5 Customer Feedback Analysis

Reviews of competing tools were analysed across the Apple App Store, Google Play Store and G2, alongside a survey of 47 Saudi marketing professionals in Riyadh and Jeddah (Q3 2026).

Positive feedback (what users value).

- All-in-one dashboards that eliminate switching between tools — cited by 68% of respondents
- AI that drafts social captions in Arabic without manual translation — 61%
- Affordable pricing in SAR that avoids foreign-exchange risk — 54%
- Automated weekly performance reports that save an average of 6 hours per week

Negative feedback and market gaps.

- Existing tools lack Arabic AI that captures Saudi colloquial dialect and cultural nuance, forcing marketers to rewrite AI output (about 4 hours per week).
- No tool integrates Snapchat, X Analytics, Instagram Insights and Google Search Console in one Arabic-language dashboard; marketers log into four platforms and consolidate data by hand.

The survey sample is small and limited to two cities, so it supports the existence of demand but not its exact size; the closed beta will test willingness to pay.

3.6 Market Opportunity and Differentiation

Tempo's advantage is the combination no competitor offers: Saudi-dialect AI, publishing to Snapchat, X and Instagram, social and website analytics in one dashboard, and SAR pricing from SAR 189. These features can be copied over time, so speed to market and Arabic content quality are what Tempo must protect.

3.7 Marketing Strategy

Positioning: "the Saudi marketer's co-pilot" — the only affordable tool that writes in Saudi Arabic, publishes to Snapchat, X and Instagram, and reports on social and website results together. Messaging stresses time saved rather than technology. **Pricing:** a free tier lowers risk for SMEs; Starter at SAR 189 is close to Jasper AI (content only) and far below Hootsuite and Sprout Social. About one in nine free users is assumed to convert to paying.

Channels and retention. Pre-launch waitlist and demos in Riyadh and Jeddah (Q4 2026), Saudi launch for SMEs and creators (Q1 2027), agency expansion (H1 2027), then a GCC pilot. Channels are Arabic content on Snapchat, Instagram and X, workshops with chambers of commerce and coworking spaces, LinkedIn agency outreach, bilingual SEO and a referral programme. Year 1 marketing budget is SAR 90,000. Retention relies on the Monday report, the content calendar and agency workspaces.

4 Feasibility Study

4.1 Summary

Tempo is **feasible on conditions**. It can be built by a small team with proven technologies and a commercial LLM API, and the market gap is well supported. The financial case depends on reaching about **614 paying accounts**, so spending is released in stages. Pre-launch costs are about SAR 115,000 and funding of SAR 1.0 million in two rounds covers the plan. Monthly break-even is expected in mid-2029.

4.2 Products and Plans

Tempo is a subscription-based, mobile-responsive web platform with: bilingual and Saudi-dialect AI content, one content calendar for Snapchat, X and Instagram, a unified social + Search Console dashboard, automated weekly bilingual reports (branded PDF for agencies), Saudi-native timing and trend insights, client workspaces and role-based team access.

Plan	SAR/mo	For	Includes
Free / trial	0	Evaluation, creators	One workspace, limited AI generations and posts; no PDF branding
Starter	189	Solo marketers, SMEs	Three platforms, Search Console, AI content, scheduling, weekly bilingual report
Agency	499	Boutique agencies	Client workspaces, team roles, branded bilingual PDFs, higher AI allowance

Table 3: Plan Structure (prices exclude 15% VAT)

4.3 Market Feasibility

Market size (SAR 3.5B TAM) and survey demand support Tempo; Year 3 revenue is only about 0.19% of the SAM (Section 3).

4.4 Technical Feasibility

The approach is to build the product-specific parts (workspaces, content studio, dashboard, reports) and buy the rest (cloud, AI models, payments). Main risks are third-party APIs and AI dialect quality, not computing power.

Area	Choice and constraint
Technology stack	Next.js + TypeScript (RTL/LTR), NestJS REST API with a job queue, PostgreSQL and Redis, hosted in a Saudi cloud region
AI	Commercial LLM API behind a provider-independent layer; Saudi-dialect prompts scored by native speakers before each release; users edit drafts before publishing
Instagram / X	Meta app review and paid, rate-limited X API: apply in Sprint 0, cache data, queue requests
Snapchat	Restricted, approval-based access: released after X and Instagram, with CSV import as fallback
Search Console	Read-only OAuth with quotas: low risk, daily data pulls
Security & privacy	Encrypted tokens, MFA, audit logs, penetration test before launch; Saudi PDPL compliance

Table 4: Technical Feasibility

4.5 Operational Feasibility

Tempo starts with three founders (product/project manager, back-end & AI developer, front-end & UX developer) plus part-time marketing (3.5 FTE in Year 1). An AI/ML engineer, a support specialist and sales staff are added only when revenue allows, reaching 8 people by Year 3. Accounting, legal and penetration testing are outsourced.

4.6 Economic Feasibility

All figures are estimates in SAR (Year 1 = 2027). Assumptions: 85% Starter / 15% Agency mix in Year 1; paying accounts of 150, 450 and 950 at the end of Years 1–3; about 1 in 9 free users converts; AI cost of SAR 16 per paying account per month.

SAR	Year 1	Year 2	Year 3
Paying accounts (year end)	150	450	950
Revenue	176,625	865,950	2,142,660
Gross profit	127,209	693,701	1,755,974
Operating expenses	(456,300)	(1,022,650)	(1,599,000)
Net profit	**(329,091)**	**(328,949)**	**153,050**
Cash at year end	166,027	245,314	414,158
Accounts needed to break even	199	414	614

Table 5: Financial Summary (start-up costs SAR 115,000; funding SAR 600,000 in Q4 2026 and SAR 400,000 in January 2028)

Gross margin is 72–82%. Losses in Years 1–2 come from spending on the team and marketing before revenue grows. The plan passes 614 accounts during Q2 of Year 3, so **monthly break-even is expected in mid-2029**. If accounts reach only 70% of plan, cash runs out around the end of Year 2, so hiring is tied to real customer numbers.

4.7 Risk Analysis

Risk	L	I	Mitigation
Platform API access denied, delayed or changed	H	H	Apply in Sprint 0; launch X and Instagram first; CSV fallback
Global tools add Arabic AI or cut prices	M	H	Focus on dialect quality, local insights, SAR pricing; release quickly
Slow sales / low free-to-paid conversion	M	H	Stage-gated spending; early agency pilots
Weak AI or dialect quality	M	H	Human review, native-speaker evaluation set, content filters
PDPL non-compliance or security incident	L	H	Saudi hosting, encryption, MFA, penetration test, legal review
Funding not secured	M	H	Spend only against gates; grants and incubators

Table 6: Main Risks (L = likelihood, I = impact)

4.8 Findings and Recommendations

We recommend an **MVP-first, stage-gated launch**. Technical and market feasibility are good; operational feasibility depends on the founders' time; financial feasibility is conditional on raising SAR 1.0M and reaching about 600 accounts.

5 Business Model Canvas

The Business Model Canvas summarises how Tempo creates, delivers and captures value. Every block is taken from the market study (Section 3) and the feasibility study (Section 4).

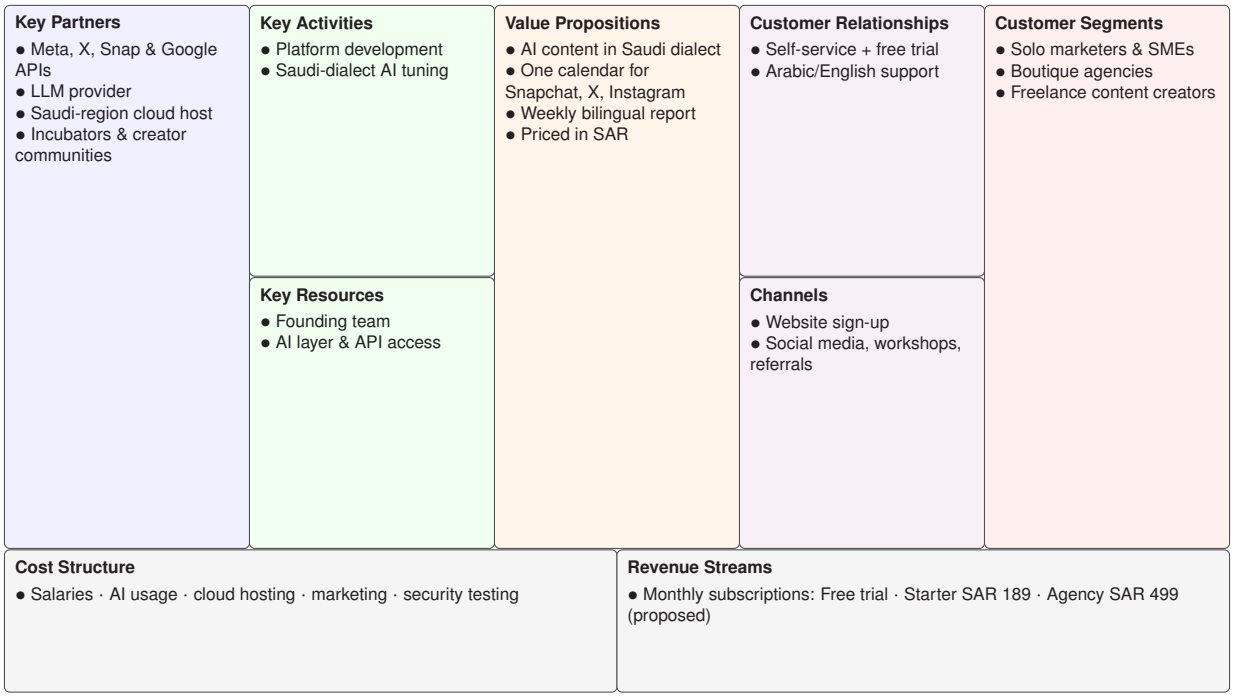


Figure 2: Tempo Business Model Canvas

6 Personas and User Stories (Lab 2)

Following a user-centred design approach, we identified Tempo's user groups, built realistic personas and mapped each persona to required features.

6.1 User Groups

User Group	Role	Frequency	Technical Level
Solo Marketers & SME Owners	Manage own social media and website marketing on Saudi platforms	Daily	Intermediate
Digital Marketing Agencies	Manage multiple client social media accounts	Daily	Intermediate–Advanced
Freelance Content Creators	Produce Arabic content for brands on Snapchat and Instagram	Several times/week	Beginner–Intermediate
Enterprise Marketing Depts.	In-house teams at large corporations needing unified analytics (deferred beyond Year 3)	Daily	Advanced

Table 7: Potential User Groups

6.2 Personas

	Nora Al-Shamrani – Solo Marketer	**Khalid Al-Otaibi** – Agency Account Manager	**Lama Al-Harbi** – Freelance Content Creator
Age, education	28; BBA Marketing, King Abdulaziz University	35; MBA, KFUPM	24; BA Graphic Design, Imam Abdulrahman Bin Faisal University
Role	Digital Marketing Manager at a Saudi e-commerce fashion brand (Riyadh)	Senior Account Manager at a boutique agency in Jeddah (8 clients)	Freelancer creating Snapchat and Instagram content for five small brands
Goals	One bilingual dashboard for Snapchat, X, Instagram; reporting from 6 h to under 1 h; Saudi-dialect AI content	All clients in one dashboard; weekly branded bilingual reports; automate content calendars	More content in less time; one calendar for all brands; simple summaries for clients
Motivations	Clear ROI reports for her manager; avoid USD-priced tools	Stand out in the Jeddah market; cut junior staff hours on reporting	Grow income and reputation; keep tool costs low
Pain points	Four platforms; English AI copy she rewrites in Saudi dialect; unpredictable USD costs	No affordable per-client workspaces; 12+ h/week compiling data; AI gives formal MSA, not Hijazi	Writes every caption from scratch for five voices; spreadsheets and reminders per brand; screenshots numbers from each app
Tech level	Intermediate	Advanced; expects team roles and permissions	Beginner–Intermediate
Usage context	Desktop in the office, phone on the move; posts 8–10 PM; needs mobile approvals	Desktop; reports every Monday; team of 4 working in different client workspaces	Mostly phone; plans on Sunday evenings on a laptop; starts on the Free plan

Table 8: User Personas

6.3 Persona-to-Requirements Mapping

Persona	Feature Needed	Priority	Req. ID
Nora (Solo Marketer)	Bilingual Arabic/English AI content drafting	High	L4-FR-B01
Nora (Solo Marketer)	One-click scheduling to Snapchat, X & Instagram	High	L4-FR-B05
Nora (Solo Marketer)	Automated weekly performance report in Arabic	High	L4-FR-D01, D02
Nora (Solo Marketer)	SAR-denominated pricing with freemium tier	High	Plans (Table 3)
Nora (Solo Marketer)	Mobile-responsive dashboard	Medium	L4-NFR-07
Khalid (Agency Manager)	Multi-account client workspace management	High	L4-FR-A03
Khalid (Agency Manager)	Branded bilingual PDF report export	High	L4-FR-D01, D02
Khalid (Agency Manager)	Unified social + website analytics (Search Console)	High	L4-FR-C06
Khalid (Agency Manager)	Team role-based access control	Medium	L4-FR-A04
Khalid (Agency Manager)	Saudi-native platform insights (Snapchat, X trends)	High	L4-FR-C04, C05
Lama (Freelance Creator)	Multiple brand voices / dialect selection in AI drafts	High	L4-FR-B01
Lama (Freelance Creator)	One calendar across several brands	Medium	L4-FR-A03, B05
Lama (Freelance Creator)	Free tier and simple upgrade path	High	Plans (Table 3)

Table 9: Persona-to-Feature Mapping

6.4 User Stories

- As a solo marketer, I want to generate Snapchat and Instagram captions in Saudi Arabic dialect so that I can connect authentically with my audience without hours of manual translation and rewriting.
- As a solo marketer, I want to view all my social media and website analytics in a single Arabic-language dashboard so that I can make data-driven decisions without logging into four platforms.
- As a solo marketer, I want to schedule one campaign to Snapchat, X and Instagram at 9 PM from one calendar so that I reach my audience at peak time without posting manually.
- As a solo marketer, I want a weekly report generated automatically in Arabic so that I can share results with my manager in two clicks.

- As an agency account manager, I want to manage all my client accounts from a single dashboard so that I can switch between client workspaces without logging out and back in.
- As an agency account manager, I want to generate branded bilingual PDF reports automatically so that I can deliver client-ready insights in under 5 minutes instead of 3 hours.
- As an agency account manager, I want to assign drafts to junior team members and approve them before publishing so that client content is always checked.
- As an agency account manager, I want to give a client view-only access so that they can see results without being able to change anything.
- As a freelance creator, I want to keep separate brand voices and calendars for each client so that I can manage five brands from one account.
- As a freelance creator, I want to start on a free plan and upgrade in SAR when I get more clients so that tool costs grow with my income.
- As a team editor, I want to be notified when my draft is approved or rejected so that I can act quickly.
- As an account owner, I want to enable multi-factor authentication and revoke connected accounts so that my brand's social accounts stay secure.

6.5 Usage Scenarios

Nora types an Eid al-Adha brief, gets three Saudi-dialect captions in 10 seconds, schedules them to Snapchat, X and Instagram for 9 PM and sends her manager a bilingual PDF report in two clicks. **Khalid** switches between eight client workspaces every Monday, generates branded reports for three clients and assigns drafts to two junior editors for approval. **Lama** generates a week of captions for five brands on Sunday evening (playful Hijazi for a café, elegant MSA for a perfume brand) and sends each client a one-page summary from her phone.

7 Requirements Elicitation Methods

Requirements were gathered from users, from the market and from technical sources, then validated through reviews. Table 10 lists each method, what the team did and which requirements it produced.

Method	What we did	Key findings	Requirements produced
Questionnaire (survey)	Survey of 47 Saudi marketing professionals in Riyadh and Jeddah, Q3 2026	68% want one dashboard; 61% want Arabic AI captions; 54% want SAR pricing	L4-FR-B01, L4-FR-C06, plan pricing
Review analysis	Reviews of Hootsuite, Buffer, Sprout Social and Jasper AI on the App Store, Google Play and G2	No Saudi dialect; no Snapchat; four logins; about 6 h/week of manual reports	L4-FR-B01, B05, L4-FR-D01
Competitor analysis	Feature and price comparison of five tools (Table 2)	No tool combines Saudi platforms, Search Console and SAR pricing	L4-FR-C01, C06
Personas & scenarios	Personas and usage scenarios mapped to features with priorities (Section 6)	Different needs for solo marketers, agencies and creators	L4-FR-A03, A04, L4-NFR-07
Peer review & consistency checklist	Team members reviewed each other's labs and checked IDs, numbers and targets across all documents	Requirements that were not SMART or not consistent between documents	Final SMART version (Section 8)

Table 10: Elicitation Methods and Their Output

Validation. Every requirement was checked against the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) and traced to a persona and a use case (Table 21). The survey sample is small (47 people in two cities), so the closed beta will confirm the findings.

8 Lab 4: Software Requirements Specification (SRS)

8.1 Introduction

This document specifies the software requirements for Tempo, an AI-powered bilingual (Arabic/English) marketing co-pilot for solo marketers, SMEs, and digital agencies in Saudi Arabia. Tempo integrates four platforms — Snapchat, X, Instagram, and Google Search Console — into a single bilingual dashboard. This document consolidates the use cases identified in Lab 3 into four simplified, high-priority scenarios and derives the functional and non-functional requirements that the system must satisfy. All requirements are traceable to the user personas (Nora Al-Shamrani and Khalid Al-Otaibi) and the use cases defined here. For the midterm, every requirement in this SRS was revised to be SMART; this is the final version of the project requirements.

8.2 Lab Objectives

1. Simplify and consolidate use cases into four core workflows.
2. Define functional requirements (FR) organised by system module.
3. Specify non-functional requirements (NFR) covering performance, security, and usability.
4. Produce a requirements traceability matrix linking personas → use cases → requirements.

8.3 Part 1: Simplified Use Cases

The six use cases from Lab 3 (Section 9) have been consolidated into four core use cases covering the primary workflows of Tempo. Each is presented in brief format (name, actor, trigger, main flow, outcome).

8.3.1 Use Case Diagram

Figure 3 shows all four simplified use cases, the primary actors (Nora and Khalid), and the external system actors they interact with.

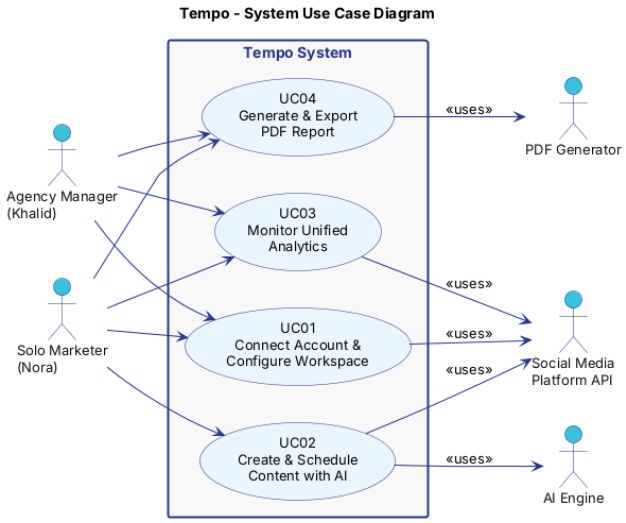


Figure 3: Lab 4 – Tempo System Use Case Diagram

8.3.2 Use Case Overview

UC	Name	Primary Actor	Trigger	Outcome
L4-UC01	Connect Account & Configure Workspace	Nora / Khalid	User signs up or logs in	Social accounts linked; workspace ready
L4-UC02	Create & Schedule Content with AI	Nora	User clicks “Generate Content”	Post scheduled on selected platforms
L4-UC03	Monitor Unified Analytics	Nora / Khalid	User opens Dashboard	Cross-platform metrics displayed; alerts triggered
L4-UC04	Generate & Export PDF Report	Khalid / Nora	User clicks “Generate Report”	Branded bilingual PDF downloaded or shared

Table 11: Lab 4 Use Case Overview

L4-UC01 – Connect Account & Configure Workspace

Field	Description
Use Case ID	L4-UC01
Use Case Name	Connect Account & Configure Workspace
Actors	Solo Marketer (Nora), Agency Manager (Khalid)
Precondition	User has a Tempo account (free or paid) and at least one social platform account.
Main Flow	1. User navigates to Settings → Connected Accounts. 2. Selects a platform (Instagram, X, Snapchat, or Google Search Console). 3. Authenticates via OAuth 2.0 and grants required permissions. 4. Tempo confirms the connection and displays the account handle / property within 10 seconds. 5. Agency user additionally creates or names a client workspace and assigns team members.
Alternative Flow	OAuth fails → error message shown; user retries or contacts support.
Postcondition	Social account linked to workspace; first data sync starts within 5 minutes.
Priority	Must Have

Table 12: L4-UC01 – Connect Account & Configure Workspace

L4-UC02 – Create & Schedule Content with AI

Field	Description
Use Case ID	L4-UC02
Use Case Name	Create & Schedule Content with AI
Actors	Solo Marketer (Nora), AI Engine, Social Media Platform API
Precondition	At least one social account is connected; user has remaining AI generation quota.
Main Flow	1. User selects platform and language (Arabic / English / Bilingual). 2. Clicks “Generate Content”; optionally adds a brief or theme. 3. AI Engine analyses the top 10 posts of the last 30 days and returns 3 content variants with engagement scores (0–100) within 10 seconds. 4. User reviews variants, optionally edits, then selects one. 5. User sets publish date/time and clicks “Schedule”. 6. Tempo queues the post and confirms the scheduled time.
Alternative Flow	Fewer than 5 past posts → AI uses brand style guide template. Platform API error → post saved as draft with error notification.
Postcondition	Post queued in content calendar; platform notified at scheduled time.
Priority	Must Have

Table 13: L4-UC02 – Create & Schedule Content with AI

Activity Diagram – L4-UC02 Figure 4 shows the detailed activity flow for L4-UC02, including the parallel AI processing steps and the decision branches for user satisfaction and publish success.

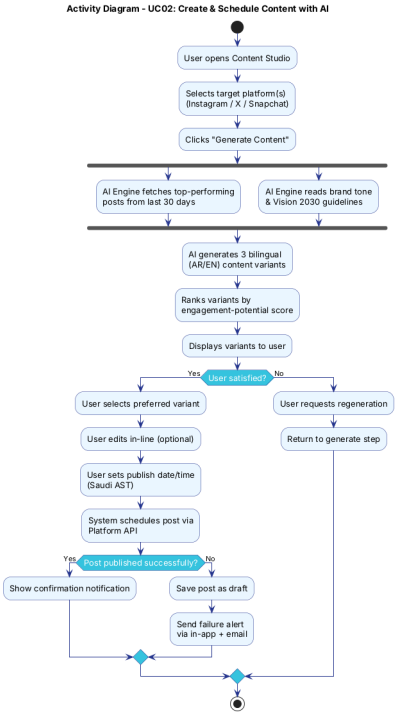


Figure 4: Activity Diagram: Create & Schedule Content with AI (L4-UC02)

L4-UC03 – Monitor Unified Analytics

Field	Description
Use Case ID	L4-UC03
Use Case Name	Monitor Unified Analytics
Actors	Solo Marketer (Nora Al-Shamrani), Agency Manager (Khalid Al-Otaibi), Analytics Service
Precondition	At least one social account and/or Google Search Console property connected; at least 24 hours of data collected.
Main Flow	1. User opens the Dashboard tab. 2. System displays cross-platform social metrics (reach, engagement rate, follower growth) alongside Google Search Console website data (clicks, impressions, CTR). 3. User selects a date range (7 days / 30 days / custom). 4. Dashboard updates the charts within 3 seconds. 5. If any metric drops more than 20% week-over-week, a performance alert is displayed.
Alternative Flow	No data for selected range → placeholder message; system prompts to connect account or wait for sync.
Postcondition	User has an up-to-date cross-platform performance overview; alerts acknowledged or dismissed.
Priority	Must Have

Table 14: L4-UC03 – Monitor Unified Analytics

L4-UC04 – Generate & Export PDF Report

Field	Description
Use Case ID	L4-UC04
Use Case Name	Generate & Export PDF Report
Actors	Agency Manager (Khalid), Solo Marketer (Nora)
Precondition	Analytics data available for the selected period; branding assets configured for agency workspace.
Main Flow	1. User clicks “Generate Report” and selects the reporting period. 2. Chooses language (Arabic / English / Both) and report type (Summary / Full). 3. System compiles reach, engagement, top posts, and platform breakdown. 4. PDF is rendered within 60 seconds with client branding and correct RTL/LTR layout. 5. User downloads the PDF or copies a share link (valid 7 days).
Alternative Flow	No data for period → report generated with a “No data available” section and a note to the user.
Postcondition	Branded bilingual PDF saved and/or shared; report log updated.
Priority	Must Have

Table 15: L4-UC04 – Generate & Export PDF Report

8.4 Part 2: Functional Requirements

Functional requirements are organised by system module. Each requirement is identified by a unique ID and linked to the use case it supports.

8.4.1 Module A – Authentication & Workspace Management

FR ID	Requirement	Linked UC	Priority
L4-FR-A01	The system shall allow users to register and log in using email/password with optional multi-factor authentication (authenticator app), and shall lock an account for 15 minutes after 5 failed login attempts.	L4-UC01	Must Have
L4-FR-A02	The system shall allow users to connect social accounts (Instagram, X, Snapchat) and Google Search Console properties via OAuth 2.0, confirm the connection within 10 seconds, and start the first data sync within 5 minutes.	L4-UC01	Must Have
L4-FR-A03	Agency users shall be able to create at least 10 client workspaces and switch between them in under 2 seconds without logging out.	L4-UC01	Must Have
L4-FR-A04	The system shall support role-based access control with roles Owner, Admin, Editor, and Viewer; 100% of actions outside a user's role shall be refused, verified by automated tests.	L4-UC01	Must Have
L4-FR-A05	Users shall be able to revoke social account connections at any time from the Settings panel; the stored tokens shall be deleted within 1 minute.	L4-UC01	Should Have

Table 16: Module A – Authentication & Workspace Management

8.4.2 Module B – AI Content Generation & Scheduling

FR ID	Requirement	Linked UC	Priority
L4-FR-B01	The system shall generate up to 3 content variants per request within 10 seconds using a commercial LLM API, in Arabic, English, or both.	L4-UC02	Must Have
L4-FR-B02	Content generation shall use the user's 10 posts with the highest engagement rate in the past 30 days as context for the prompt (or the brand style template if there are fewer than 5 posts).	L4-UC02	Must Have
L4-FR-B03	Generated content shall include an engagement-potential score from 0 to 100 for each variant, ranked highest first.	L4-UC02	Should Have
L4-FR-B04	Users shall be able to edit any generated variant in-line before scheduling, with a live character count against each platform's limit.	L4-UC02	Must Have
L4-FR-B05	The system shall schedule posts to Instagram, X, and Snapchat at a user-specified date and time (Saudi AST), to the minute, up to 90 days ahead.	L4-UC02	Must Have
L4-FR-B06	If a scheduled post fails to publish after 3 retries, the system shall notify the user by in-app alert and email within 2 minutes, and save the post as a draft.	L4-UC02	Must Have

Table 17: Module B – AI Content Generation & Scheduling

8.4.3 Module C – Unified Analytics Dashboard

FR ID	Requirement	Linked UC	Priority
L4-FR-C01	The system shall display a unified dashboard showing reach, engagement rate, and follower growth for all connected social platforms (Snapchat, X, Instagram).	L4-UC03	Must Have
L4-FR-C02	The dashboard shall support date-range filtering: last 7 days, last 30 days, and a custom range of up to 12 months.	L4-UC03	Must Have
L4-FR-C03	The system shall collect platform metrics automatically via scheduled API calls once every 24 hours, completed by 06:00 AST.	L4-UC03	Must Have
L4-FR-C04	The system shall display a "Top Performing Post" card per platform, updated daily by 07:00 AST.	L4-UC03	Should Have
L4-FR-C05	The system shall trigger a performance alert within 1 hour of the daily sync when any metric drops more than 20% week-over-week.	L4-UC03	Should Have
L4-FR-C06	The system shall display Google Search Console data (queries, clicks, impressions, and average position) alongside social metrics in the unified dashboard.	L4-UC03	Must Have

Table 18: Module C – Unified Analytics Dashboard

8.4.4 Module D – PDF Report Generation

FR ID	Requirement	Linked UC	Priority
L4-FR-D01	The system shall generate a PDF performance report for any selected date range of up to 12 months within 60 seconds of the user's request.	L4-UC04	Must Have
L4-FR-D02	Reports shall be available in Arabic (RTL layout), English (LTR layout), or both languages in a single document.	L4-UC04	Must Have
L4-FR-D03	Agency workspace reports shall include the client logo and two brand colours configured by the agency.	L4-UC04	Must Have
L4-FR-D04	Reports shall include: Executive Summary, Platform Breakdown, Top 5 Posts, and at least 3 Recommendations.	L4-UC04	Should Have
L4-FR-D05	The system shall allow reports to be downloaded as PDF or shared via a secure link valid for 7 days.	L4-UC04	Should Have

Table 19: Module D – PDF Report Generation

8.5 Part 3: Non-Functional Requirements

Non-functional requirements define the quality attributes the system must exhibit regardless of specific functionality. They are organised by category.

NFR ID	Category	Requirement	Priority
L4-NFR-01	Performance	The dashboard shall load and display all platform metrics within 3 seconds (95th percentile) on a 4G or broadband connection.	Must Have
L4-NFR-02	Performance	AI content generation shall return results within 10 seconds (95th percentile) for a standard request (3 variants, 1 platform).	Must Have
L4-NFR-03	Availability	The system shall maintain 99.5% uptime during Saudi peak hours (6 AM – 12 AM AST, Saturday – Thursday).	Must Have
L4-NFR-04	Security	All social account OAuth tokens shall be encrypted at rest using AES-256 and transmitted only over HTTPS (TLS 1.2+).	Must Have
L4-NFR-05	Security	The system shall undergo an external penetration test before the public launch and remediate all Critical and High findings.	Must Have
L4-NFR-06	Privacy	The system shall comply with the Saudi PDPL: user data stored in a Saudi-region cloud, deletable on request within 30 days.	Must Have
L4-NFR-07	Usability	The interface shall support right-to-left (RTL) Arabic text alongside left-to-right (LTR) English on 100% of screens, checked against a UI checklist before every release.	Must Have
L4-NFR-08	Usability	At least 80% of closed-beta testers shall connect a social account and generate their first content suggestion within 10 minutes of sign-up.	Should Have
L4-NFR-09	Localisation	All date, time, currency, and calendar displays shall default to Saudi Arabia conventions (AST, SAR, Hijri/Gregorian toggle).	Should Have
L4-NFR-10	Scalability	The architecture shall support horizontal scaling to serve at least 5,000 concurrent users without degrading below L4-NFR-01/L4-NFR-02 thresholds, verified by load test before January 2028.	Should Have
L4-NFR-11	Maintainability	The AI content module shall be provider-independent, allowing the LLM provider to be swapped in at most 5 working days without changes to other modules.	Should Have

Table 20: Lab 4 Non-Functional Requirements

8.6 Part 4: Requirements Traceability Matrix

The matrix below links each persona to the use case that represents their primary goal, and maps the functional requirements that implement it.

Persona	Use Case	Functional Requirements	NFRs
Nora (Solo Marketer)	L4-UC01: Connect Account	L4-FR-A01, L4-FR-A02, L4-FR-A05	L4-NFR-04, L4-NFR-07
Nora	L4-UC02: Create & Schedule Content	L4-FR-B01, L4-FR-B02, L4-FR-B03, L4-FR-B04, L4-FR-B05, L4-FR-B06	L4-NFR-01, L4-NFR-02, L4-NFR-07, L4-NFR-08
Nora / Khalid	L4-UC03: Monitor Analytics	L4-FR-C01, L4-FR-C02, L4-FR-C03, L4-FR-C04, L4-FR-C05, L4-FR-C06	L4-NFR-01, L4-NFR-03, L4-NFR-07
Khalid (Agency Manager)	L4-UC01: Configure Workspace	L4-FR-A03, L4-FR-A04, L4-FR-A05	L4-NFR-04, L4-NFR-06, L4-NFR-10
Khalid	L4-UC04: Generate PDF Report	L4-FR-D01, L4-FR-D02, L4-FR-D03, L4-FR-D04, L4-FR-D05	L4-NFR-06, L4-NFR-07, L4-NFR-09

Table 21: Lab 4 Requirements Traceability Matrix

9 Use Cases and UML (Lab 3)

9.1 Actors

- **Primary:** Marketer (solo marketer, SME owner or freelance creator), Agency Manager (workspace Owner/Admin), Team Member (Editor or Viewer, including a client given view-only access)
- **Secondary:** LLM Provider (AI API), Social Platforms (Instagram, X, Snapchat APIs), Google Search Console, Payment Gateway, Scheduler (system timer that triggers publishing, data sync and weekly reports)

9.2 Use Case Titles

ID	Use case title	Primary actor	Priority	Requirements (Lab 4)
UC-01	Generate AI Content	Marketer / Team Editor	Critical	L4-FR-B01–B04, L4-NFR-02, L4-NFR-11
UC-02	Schedule & Publish Posts (extended by *Approve Post*)	Marketer / Team Editor	Critical	L4-FR-B05, B06, A04, L4-NFR-03
UC-03	View Unified Analytics	Marketer / Agency Manager / Viewer	High	L4-FR-C01–C06, L4-NFR-01, L4-NFR-03
UC-04	Generate Bilingual Report	Marketer / Agency Manager	High	L4-FR-D01–D05, L4-NFR-07, L4-NFR-09
UC-05	Manage Client Workspaces & Team Roles	Agency Manager	Medium	L4-FR-A03, A04, L4-NFR-04
UC-06	Manage Account, Security & Subscription	User	Critical	L4-FR-A01, A02, A05, L4-NFR-04, L4-NFR-06

Table 22: Use Case Titles

Lab 4 (Section 8) consolidates these into four simplified use cases: L4-UC01 Connect Account & Configure Workspace, L4-UC02 Create & Schedule Content with AI, L4-UC03 Monitor Unified Analytics, and L4-UC04 Generate & Export PDF Report.

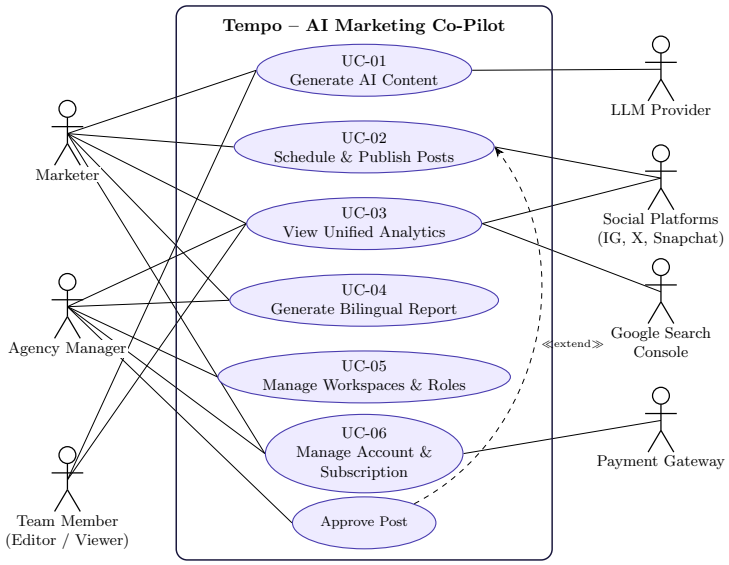


Figure 5: System Use Case Diagram

9.3 Detailed Use Cases

9.3.1 UC-01: Generate AI Content

Use Case	UC-01 Generate AI Content
Actors	Marketer or Team Editor (primary); LLM Provider (secondary)
Preconditions	User is logged in, has a workspace and remaining AI credits for the plan.
Trigger	User clicks “Generate” in the AI Content Studio.
Main Flow	1. User opens the AI Content Studio in the active workspace. 2. User enters a brief (up to 500 characters) and selects dialect (Najdi, Hijazi, MSA, English), platforms, tone and occasion. 3. System validates the brief and checks the AI credit balance. 4. System builds a dialect-specific prompt and sends it to the LLM Provider. 5. LLM Provider returns generated text; system applies content filters and platform length limits. 6. System displays three drafts within 10 seconds, with character counts and hashtag/timing suggestions. 7. User edits, rates (1–5 stars) or regenerates a draft. 8. User saves the draft or adds it to the calendar (extends to UC-02).
Alternative Flows	3a. Credits exhausted: system shows upgrade options (UC-06). 4a. Provider does not answer within 10 s: system retries once with the fallback provider, otherwise shows an error and keeps the brief. 5a. Filter flags unsuitable content: the draft is withheld and regenerated.
Postconditions	Brief and drafts are stored in the workspace; AI usage is recorded.
Requirements	L4-FR-B01–B04, L4-NFR-02, L4-NFR-07, L4-NFR-11

Table 23: Use Case Description: UC-01 Generate AI Content

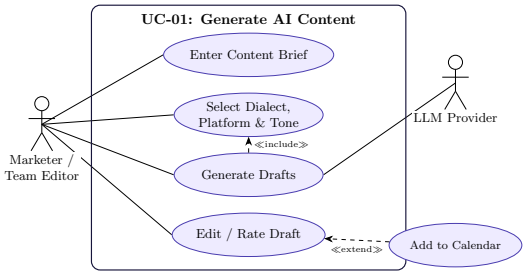


Figure 6: UC-01 Use Case Diagram

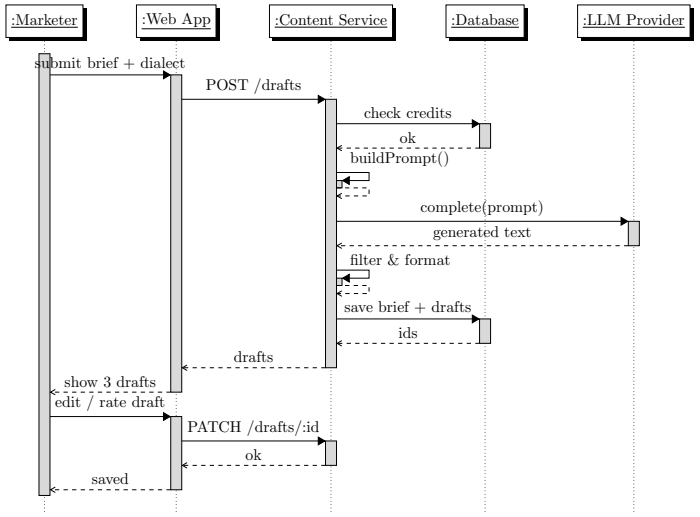


Figure 7: UC-01 Sequence Diagram: Generate AI Content

9.3.2 UC-02: Schedule & Publish Posts

Use Case	UC-02 Schedule & Publish Posts
Actors	Marketer / Team Editor (primary); Agency Manager as approver; Social Platforms and Scheduler (secondary)
Preconditions	A draft exists; at least one social account is connected with a valid token.
Trigger	User drags a draft onto the calendar or clicks “Schedule”.
Main Flow	1. User selects platforms and a date/time; system highlights the three best slots of the week (usually 8–10 PM). 2. System validates connected accounts, media format and caption length per platform. 3. If the workspace requires approval, system notifies the approver, who approves the post. 4. System saves the post as scheduled and adds a job to the publishing queue. 5. At the scheduled time the Scheduler triggers the worker. 6. Worker publishes through the platform’s API and receives the external post ID. 7. System marks the post published and notifies the user, starts metric collection and writes the audit log.
Alternative Flows	2a. Token expired: system asks the user to reconnect the account. 3a. Approver rejects: draft returns to the editor with comments. 6a. API error or rate limit: worker retries 3 times with back-off, then marks failed and alerts the user within 2 minutes. 6b. Snapchat not yet approved: system offers “export and post manually”.
Postconditions	Post is live on the platform(s) and its status is recorded; metric collection is scheduled.
Requirements	L4-FR-A04, L4-FR-B05, L4-FR-B06, L4-NFR-03

Table 24: Use Case Description: UC-02 Schedule & Publish Posts

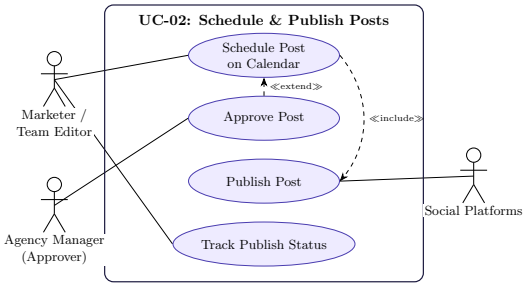


Figure 8: UC-02 Use Case Diagram

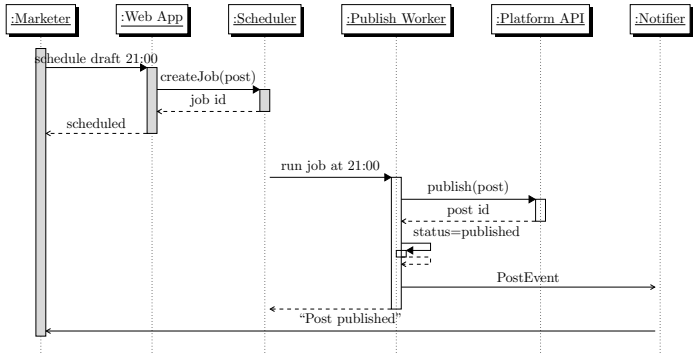


Figure 9: UC-02 Sequence Diagram: Schedule & Publish Posts

9.3.3 UC-03: View Unified Analytics

Use Case	UC-03 View Unified Analytics
Actors	Marketer, Agency Manager or Viewer (primary); Social Platforms, Google Search Console, Scheduler (secondary)
Preconditions	Accounts and website property are connected; at least one day of data has been collected.
Trigger	User opens the Dashboard or Analytics page.
Main Flow	1. Every day at 06:00 (Saudi time) the Scheduler runs collectors that pull metrics from each platform and Search Console. 2. System stores daily metrics per account and refreshes aggregate views. 3. User opens the dashboard and selects a date range and platforms. 4. System aggregates reach, engagement, followers, clicks, impressions and queries. 5. System computes Saudi-native insights (best times to post, rising queries, upcoming occasions). 6. System displays KPI cards, charts, top posts, search queries and insights in the chosen language within 3 seconds.
Alternative Flows	1a. A source is disconnected or over quota: system shows last sync time and a reconnect banner. 3a. No data yet: system shows an onboarding empty state. 6a. User exports the data as CSV.
Postconditions	User sees one up-to-date view of social and website performance.
Requirements	L4-FR-C01–C06, L4-NFR-01, L4-NFR-03

Table 25: Use Case Description: UC-03 View Unified Analytics

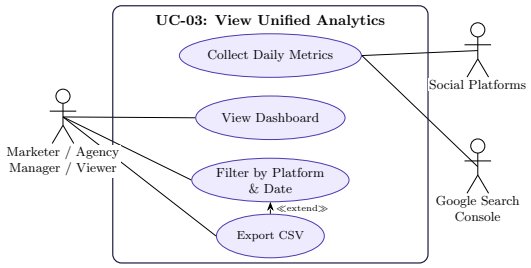


Figure 10: UC-03 Use Case Diagram

Use Case	UC-03 View Unified Analytics
Actors	Marketer, Agency Manager or Viewer (primary); Social Platforms, Google Search Console, Scheduler (secondary)
Preconditions	Accounts and website property are connected; at least one day of data has been collected.
Trigger	User opens the Dashboard or Analytics page.
Main Flow	1. Every day the Scheduler runs collectors that pull metrics from each platform and Search Console. 2. System stores daily metrics per account and refreshes aggregate views. 3. User opens the dashboard and selects a date range and platforms. 4. System aggregates reach, engagement, followers, clicks, impressions and queries. 5. System computes Saudi-native insights (best time to post, trending queries, upcoming occasions). 6. System displays KPI cards, charts, top posts, search queries and insights in the chosen language.
Alternative Flows	1a. A source is disconnected or over quota: system shows last sync time and a reconnect banner. 3a. No data yet: system shows an onboarding empty state. 6a. User exports the data as CSV.
Postconditions	User sees one up-to-date view of social and website performance.
Requirements	RQ-F2, RQ-F6, RQ-F7, RQ-F13, RQ-NF1, RQ-NF2, RQ-NF5, RQ-NF6, RQ-NF7

Figure 11: UC-03 Sequence Diagram: View Unified Analytics

9.3.4 UC-04: Generate Bilingual Report

Use Case	UC-04 Generate Bilingual Report
Actors	Marketer or Agency Manager (primary); Scheduler, Client/Viewer (secondary)
Preconditions	Workspace has collected metrics for the selected period.
Trigger	User clicks “Generate PDF report”, or the Scheduler triggers the Monday 09:00 auto-report.
Main Flow	1. User selects workspace, period, language (Arabic, English or both) and sections. 2. System loads aggregated metrics and top posts for the period. 3. System asks the AI service for a short executive summary in each selected language. 4. System fills the report template (RTL for Arabic) and applies client branding if enabled. 5. System renders the PDF with Arabic-capable fonts and stores it. 6. System notifies the user and shares a link or emails the PDF to the client.
Alternative Flows	1a. Branding requested on Starter plan: system offers the Agency upgrade. 2a. A source has missing days: report marks the gap instead of failing.
Postconditions	A bilingual PDF report is stored in the workspace and delivered.
Requirements	L4-FR-D01–D05, L4-NFR-07, L4-NFR-09

Table 26: Use Case Description: UC-04 Generate Bilingual Report

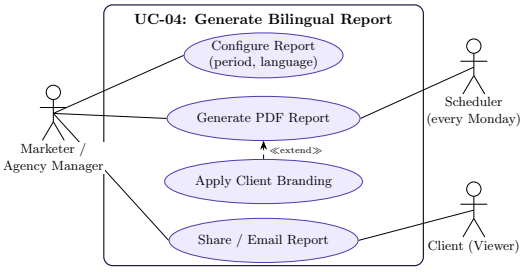


Figure 12: UC-04 Use Case Diagram

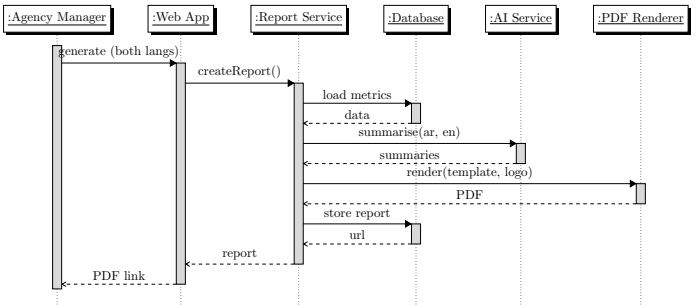


Figure 13: UC-04 Sequence Diagram: Generate Bilingual Report

9.3.5 UC-05: Manage Client Workspaces & Team Roles

Use Case	UC-05 Manage Client Workspaces & Team Roles
Actors	Agency Manager – Owner/Admin (primary); Team Member (secondary)
Preconditions	User is on the Agency plan and has the Owner or Admin role.
Trigger	User clicks “New client” or “Invite member”.
Main Flow	1. Agency Manager creates a client workspace (name, logo, default language). 2. Agency Manager connects the client’s social accounts and website (UC-06 / L4-FR-A02). 3. Agency Manager invites a team member by email and assigns a role (Admin, Editor or Viewer) per workspace. 4. System sends the invitation; the member accepts and sets a password. 5. System applies permissions; the member sees only the allowed workspaces. 6. Users switch between allowed workspaces from the top bar in under 2 seconds without logging out.
Alternative Flows	1a. Workspace limit of the plan reached: system offers an upgrade. 4a. Invitation expires after 7 days: Admin can resend it.
Postconditions	Workspace and memberships are stored; access is enforced on every request.
Requirements	L4-FR-A03, L4-FR-A04, L4-NFR-04

Table 27: Use Case Description: UC-05 Manage Client Workspaces & Team Roles

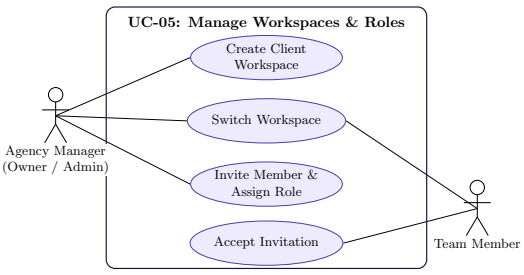


Figure 14: UC-05 Use Case Diagram

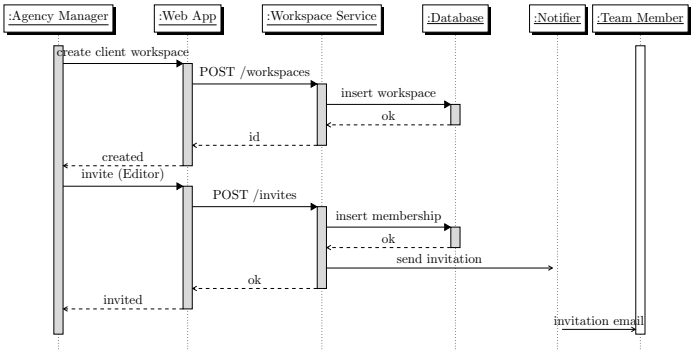


Figure 15: UC-05 Sequence Diagram: Manage Workspaces & Roles

9.3.6 UC-06: Manage Account, Security & Subscription

Use Case	UC-06 Manage Account, Security & Subscription
Actors	User (primary); Payment Gateway, Social Platforms (secondary)
Preconditions	None for registration; logged in for other steps.
Trigger	User signs up, opens Settings, or selects a plan.
Main Flow	1. User registers with name, email, password, user type and interface language. 2. System verifies the email (link valid 24 hours) and creates a personal workspace. 3. User optionally enables MFA (authenticator app). 4. User chooses a plan; system sends the SAR amount plus 15% VAT to the Payment Gateway. 5. Payment Gateway confirms payment; system activates the subscription and issues a VAT invoice within 1 minute. 6. User connects social accounts through OAuth; system stores encrypted tokens with minimum scopes.
Alternative Flows	3a. Lost MFA device: user signs in with a recovery code. 4a. Payment declined: system keeps the Free plan and shows the gateway message. 6a. User revokes a connection: tokens are deleted within 1 minute.
Postconditions	Account, subscription and connections are active and secured.
Requirements	L4-FR-A01, L4-FR-A02, L4-FR-A05, L4-NFR-04, L4-NFR-06

Table 28: Use Case Description: UC-06 Manage Account, Security & Subscription

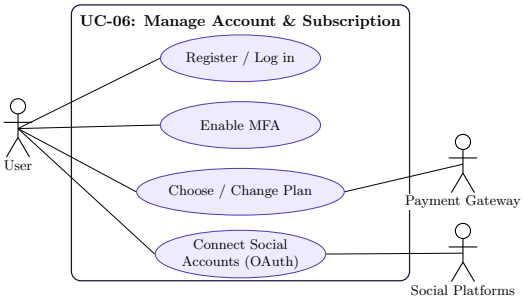


Figure 16: UC-06 Use Case Diagram

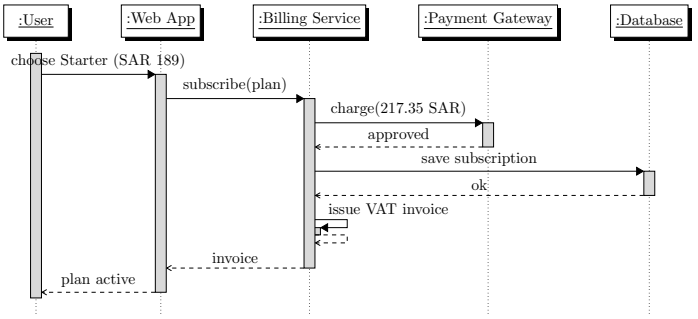


Figure 17: UC-06 Sequence Diagram: Subscribe to a Plan

10 Context Diagram

The context diagram shows Tempo as one system and the data it exchanges with its users (solo marketer and agency manager) and the external systems it depends on.



Figure 18: Tempo System Context Diagram

External entity	Data exchanged with Tempo
Solo Marketer (Nora)	Sends briefs, edits and schedules; receives drafts, dashboard and reports
Agency Manager (Khalid & team)	Sends workspaces, roles and branding; receives client reports and alerts
Social Media APIs (Instagram, X, Snapchat)	Tempo sends posts; receives metrics
Google Search Console	Tempo receives clicks and impressions
LLM Provider	Tempo sends prompts; receives drafts
Payment Gateway	Charges and payment status for subscriptions
Email Service	Tempo sends alerts and report links

Table 29: Context Diagram Data Flows

11 Project Management

11.1 Phase 1 Plan and Responsibilities

The analysis phase was planned in weekly steps: one lab per week, followed by the midterm report and presentation in weeks 5–6.



Figure 19: Phase 1 Schedule and Responsibilities

11.2 Methodology

Tempo is developed with **Agile Scrum** in two-week sprints. The Product/Project Manager acts as Product Owner, the backlog is built from the user stories (Section 6), and each sprint ends with a demo and a retrospective. Platform API access is requested in the first weeks because approvals take time; if Snapchat approval or the beta runs late, the launch *scope* is reduced (X and Instagram first) rather than the *date* moved.

11.3 Product Development Timeline

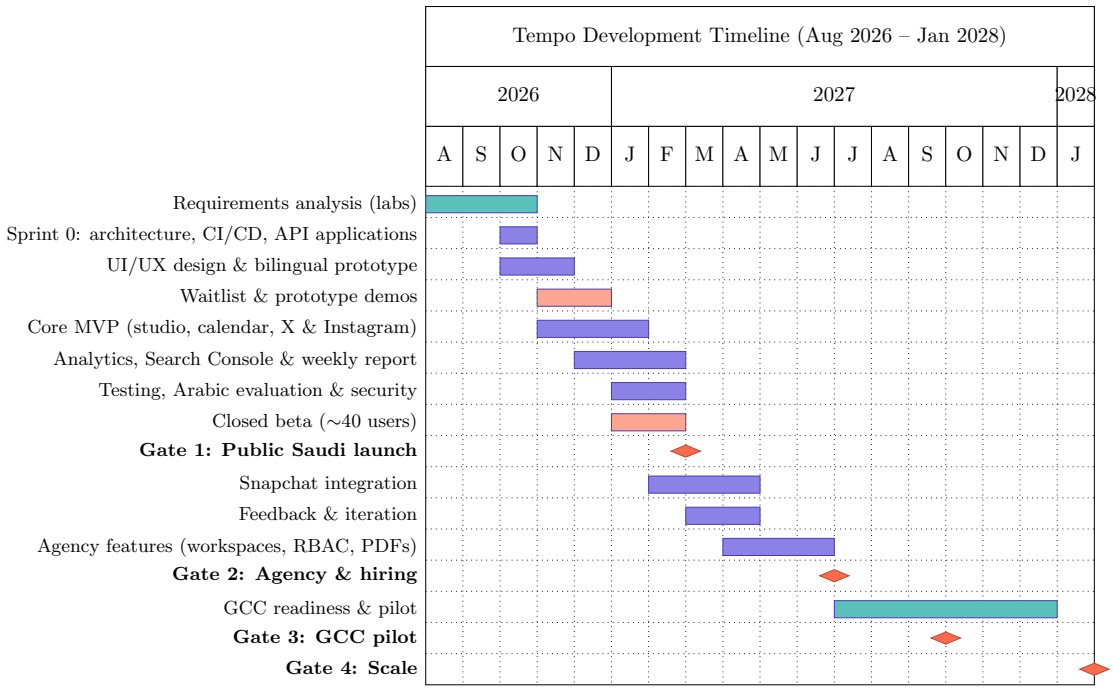


Figure 20: Product Development Gantt Chart (Aug 2026 – Jan 2028)

12 Conclusion

Tempo provides a focused solution to a clear problem: Saudi marketers lose hours every week rewriting English-first AI output, switching between four platforms and building reports by hand. By combining Saudi-dialect AI content, scheduling for Snapchat, X and Instagram, unified social and website analytics and automated bilingual reports at an SME price in SAR, Tempo fills a gap no incumbent covers. The analysis phase has delivered:

- A clear problem statement and strong market potential with competitive differentiation (SAR 3.5B TAM, 68% demand for all-in-one dashboards)
- A feasibility study showing technical and operational feasibility, and conditional economic feasibility with break-even at about 614 paying accounts in mid-2029
- A business model canvas linking segments, value, channels, costs and revenue
- Personas and user stories, and the elicitation methods used to gather requirements
- The final SMART requirements (Lab 4 SRS): 22 functional and 11 non-functional requirements
- Six use cases with UML diagrams, a context diagram, and the project plan

The next phase moves to system design (architecture, data model and user interface) and to validating demand and Arabic output quality with the prototype and the closed beta.

References

[1] Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson.

[2] Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill.

[3] Osterwalder, A., & Pigneur, Y. (2010). *Business Model Generation*. Wiley.

[4] Kingdom of Saudi Arabia. (2016). *Saudi Vision 2030*. https://www.vision2030.gov.sa

[5] Saudi Data & AI Authority (SDAIA). (2023). *Personal Data Protection Law and Implementing Regulations*. Kingdom of Saudi Arabia.

[6] United Nations. (2015). *Transforming Our World: The 2030 Agenda for Sustainable Development (SDGs)*. https://sdgs.un.org/goa

[7] Meta Platforms. (2026). *Instagram Graph API – Content Publishing and Insights*. Meta for Developers.

[8] X Corp. (2026). *X API v2 Documentation*. https://developer.x.com

[9] Snap Inc. (2026). *Snap Developer Documentation*. https://developers.snap.com

[10] Google. (2026). *Search Console API Reference*. Google for Developers.

[11] Schwaber, K., & Sutherland, J. (2020). *The Scrum Guide*. https://scrumguides.org

[12] Tempo Project Team. (2026). *Tempo Market Study*, *Lab 2: User Personas* and *Tempo Feasibility Study*. CS3151 coursework, Effat University.